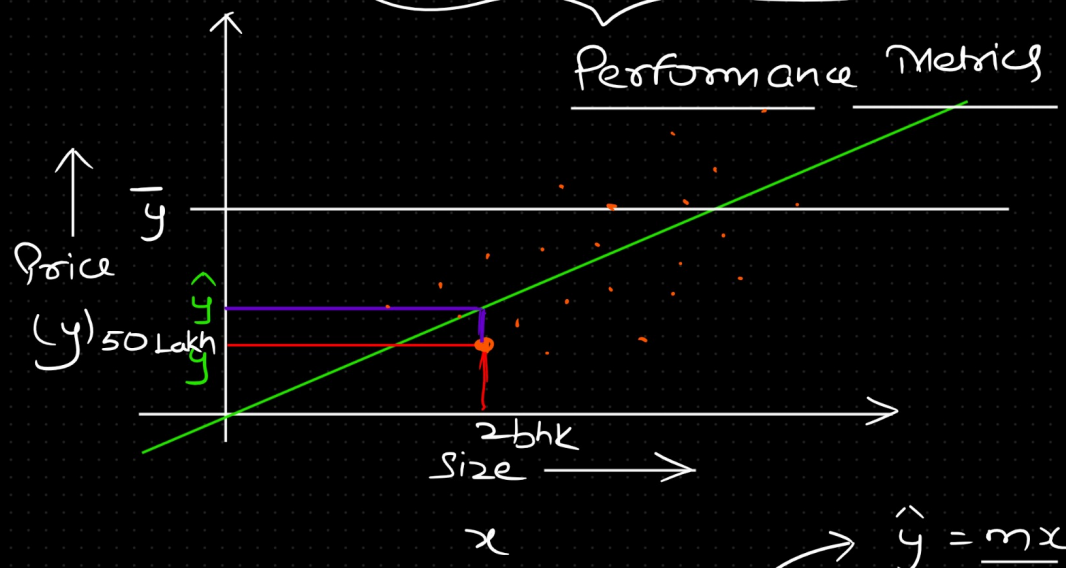
R-square vs Adjusted R-square

Performance Metrics



$\hat{y} = mx + c$

Residual $\rightarrow \sum_{i=1}^{n} (y - \hat{y}) \downarrow\downarrow$

(Error term)

$$R^2 = 1 - \frac{\text{Sum of squared error}}{\text{Total sum of square error}}$$

0 to 1

$$R^2 \Rightarrow 1 - \frac{\sum_{i=1}^{n} (y - \hat{y})^2 \downarrow\downarrow\uparrow\uparrow}{\sum_{i=1}^{n} (y - \bar{y})^2}$$

smaller

bigger

smaller R^2

Never expect

> 0.7 or 0.8

Expectations

Adjusted R-square

$$\text{Adjusted } R^2 = 1 - \frac{(1-R^2)(N-1)}{N-P-1}$$

N: Num of data points

P: Independent predictors

x_1 size, x_2 location, x_3 bedrooms, x_4 washrooms

$\downarrow$

Price of home

$\downarrow$

$\uparrow\uparrow R^2$ (0.65 $\rightarrow$ 0.85)

Adjusted $R^2 \uparrow\uparrow$

$x_5 \longrightarrow$ Gender

$\longrightarrow R^2 \uparrow\uparrow$

Adjusted $R^2 \downarrow\downarrow$